

BE/APh161: Physical Biology of the Cell
Homework 3
Due Date: Wednesday, January 28, 2026
Solutions: Sara Mahdavi

“Champions aren’t made in gyms. Champions are made from something they have deep inside them - a desire, a dream, a vision. They have to have the skill, and the will. But the will must be stronger than the skill.” - Muhammad Ali

1'. *Stuff(t)* and the Science and Mathematics of Change

Obviously, I am intensely dedicated to this idea that much of what different parts of science have focused on is the science of change. In the world of geology, huge emphasis was placed on the development of our modern notion of “Deep Time.” In physics, over and over again, we learn about dynamical laws, whether the original insights of $F = ma$ or Maxwell’s dynamical theory of the electromagnetic field, or the heat equation, etc. In chemistry, we have a beautiful and well-crafted theory of chemical reaction dynamics based upon the law of mass action.

(a) Give three one sentence examples of some time dependent phenomenon that you find interesting that qualify as examples of *stuff(t)*.

(b) Explain in a few sentences what I mean by the update rule paradigm mathematically. Show how to write an update rule for some dynamical problem.

Solution:

Update rules help you express a quantity evaluated at a given time + a small increment ($t + \Delta t$), as a function of the same quantity evaluated at previous times. We mostly work with Markovian system, meaning that the quantity at time $t + \Delta t$, only depends on the quantity at time t . We then write the quantity evaluated at time $t + \Delta t =$ the quantity at time t + the sources of variation that can happen during Δt .

1. Post-Translational Modifications and “nature’s escape from genetic imprisonment”

In a very interesting article (“Post-translational modification: nature’s escape from genetic imprisonment and the basis for dynamic information encoding”), Prof. Jeremy Gunawardena discusses how we should think about post-translational modifications as a way of expanding the natural repertoire of the 20-letter amino acid alphabet. Similarly, Prof. Christopher Walsh wrote a whole book entitled “Posttranslational Modifications of Proteins: Expanding

Nature's Inventory," again making the point that by adding chemical groups to proteins we can significantly change their properties.

(a) Provide at least one mechanistic idea about how adding a chemical group to a protein can alter its structure or function. Your answer should be offered in less than a paragraph, but should be concrete in its assertions about how these modifications change the protein. Why does Gunawardena refer to this process of post-translational modification as "escape from genetic imprisonment"?

Solutions:

An example is phosphorylation (or unphosphorylation). This adds as negatively charged phosphate group. These electrostatic interactions can change the conformational equilibrium. Although the number of possible amino acid sequences of length N is 20^N , a given organism expresses only a tiny subset encoded in its genome. Post-translational modifications expand the functional repertoire of these proteins without altering the DNA sequence, allowing rapid and reversible diversification on physiological timescales.

(b) As a toy model of the combinatorial complexity offered by post-translational modifications, imagine that a protein has N residues that are able to be phosphorylated. NOTE: comment on which residues these are and how the dominant phosphorylation chemistry differs between bacteria and eukaryotes. How many distinct proteins can be generated as a result of the N residues that can be phosphorylated? Make an approximate estimate of the mass associated with a phosphate group and what fraction of the total mass this group represents for a typical protein. Similarly, give some indication of the net charge introduced by a phosphate group at physiological pH. What ideas do you have about how we can go about measuring these different states of phosphorylation?

Solutions:

Which residues can be phosphorylated? Only hydroxyl-containing amino acids: serine, threonine, tyrosine for eukaryotes. In bacteria, we have "Reversible phosphorylation on histidine and aspartate [...], which is abundant in eubacteria", as mentioned in the paper we are studying.

How many distinct phosphorylation states are possible? For N residues, each of them can either be phosphorylated or not be phosphorylated. So two choices. So in the end, we have 2^N possibilities.

Mass added by a phosphate group A phosphate group is PO_4^{3-} , we neglect the hydrogens. We saw in previous problem set that the mass of an average protein is around $5 \times 10^4 \text{Da}$. We

know that the an oxygen atom is 16 Da. We take the phosphorus atom, to be approximately double. This gives us around 100 Da. So we see that this around 0.2% is of the total mass.

Net charge introduced at physiological pH From Bionumbers 106514, "The phosphate ion has 3 pKa values (2.1, 7.2, and 12.7) and can exist in 4 different states in aqueous solution (PO_4^{3-} , HPO_4^{2-} , H_2PO_4^- and H_3PO_4). At neutral pH, only HPO_4^{2-} (62%) and H_2PO_4^- (38%) are present in significant amounts."

How can we measure different phosphorylation states? In 2D gel electrophoresis, proteins are first separated by charge and then by size; since phosphorylation adds negative charge but barely changes mass, phosphorylated forms appear as a row of spots at the same size but shifted in charge.

(c) In this part of the problem, make a crude estimate of the number of sites on a protein that are subject to phosphorylation. To do so, imagine that the protein is a sphere with N residues packed at roughly constant density as shown in Figure ???. How does the radius of that sphere depend upon the number of residues in the protein? Given that estimate, what is the scaling of the number of residues that are on the surface with N ? Given that number, what fraction of those are phosphorylatable? Remember, these are crude estimates. Work out these results for a concrete case of a typical protein with roughly 400 amino acids.

Solutions:

Radius vs number of proteins We have the number of residues that scales like the total volume of the protein. So, we have $V \sim R^3 \sim N$, with R the radius of the protein.

Number of surface residues The surface scale like $A \sim R^2 \sim N^{2/3}$. The fraction of surface residue is $N^{-1/3}$. The fraction of phosphorylatable residues is this number times the fraction of Ser/Thr/Tyr combined. In the paper we are studying, it is mentioned that phosphorylation in eukaryotes occurs predominantly on serine and threonine. Serine abundance is 4% (Bionumbers 100640). Threonine abundance is 5% (Bionumbers 100641). So $m = N^{2/3}/10$ is our final crude estimate of the number of phosphorylatable residues.

N=400 This gives us approximately $m = 5$.

(d) Let's close out these estimates by thinking about a bacterial cell. If all 3×10^6 proteins in such a cell can be phosphorylated with the number of different phosphorylation states that you estimated above, how many distinct cells could we make with all of these different states of phosphorylation? State clearly any independence assumptions you are making when you combine phosphorylation states across proteins.

Solutions:

Assumptions All sites act independently within a protein. All proteins act independently across the cell. We treat each protein molecule individually rather than grouping by species.

Computation For m phosphorylation sites, we get 2^m possibilities. This gives, in total $2^m \times 3 \times 10^6$, even for $m = 5$ this is vastly larger than the number of atoms in the universe, so we would need to be more careful on the assumptions.

2. Energy and Life

One of the strongest things we can say about the properties of living organisms that distinguish them from inorganic materials such as the rocks that make up the face of Half Dome is that they are always consuming energy. Figure ?? shows a number of biological processes as viewed through the prism of energy consumption.

(A) Write a brief, thoughtful paragraph about the meaning of the energy scale $k_B T$. This is one of those problems for which no AI is permitted.

(B) In this problem, choose three of the entries in the figure and make your own calculation of the relevant energy scale and see to what extent you agree with the reported numbers. Don't find a way to get the same numbers as are in the figure. Rather, do this yourself and get your own number. Make sure you carefully report your thought process and assumptions.

3. Computing With the Update Rule

In class, we discussed the rate equation protocol shown in Figure ??. We used that protocol to work out the temporal dynamics of the average number of mRNAs in a population of cells assuming that the promoter of interest was constitutively expressed. In this problem, we are going to take an opportunity to compare analytic and numerical calculations of the behavior not only of the constitutive promoter, but also, of the simplest genetic switch.

(a) Go through all of the steps of the rate equation protocol shown in Figure ??(A) by explaining what the dynamical variables are, showing how to write down the update rule, reinterpreting that update rule as a differential equation and then demonstrating that the solution to that differential equation is

$$m(t) = \frac{r}{\gamma} - \left(\frac{r}{\gamma} - m_0 \right) e^{-\gamma t} \quad (1)$$

as shown in the figure. Make sure you explain the $t \rightarrow 0$ and $t \rightarrow \infty$ limits.

We have a fixed rate of production r . That is to say in the small time interval Δt , we have $r\Delta t$ molecules produced. The total number of degraded molecules is proportional to the number of molecules. Therefore we take a degradation rate per molecule number to be γ , therefore the number of molecules degraded during timescale Δt , is $\Delta t \gamma m(t)$. Putting these all together gives us,

$$m(t + \Delta t) = m(t) + r\Delta t - \gamma\Delta t m(t). \quad (2)$$

Subtracting both sides by $m(t)$ and dividing by Δt , we get

$$\frac{m(t + \Delta t) - m(t)}{\Delta t} = r - \gamma m(t). \quad (3)$$

In the limit $\Delta t \rightarrow 0$, we get,

$$\lim_{t \rightarrow 0} \frac{m(t + \Delta t) - m(t)}{\Delta t} = \frac{dm}{dt}. \quad (4)$$

This leads to the following differential equation

$$\frac{dm}{dt} + \gamma m(t) = r. \quad (5)$$

We can solve the homogeneous solution of this differential equation,

$$m_h(t) = Ae^{-\gamma t}, \quad (6)$$

with A a real constant. A particular solution is a constant m_p , satisfying the following equation

$$\frac{d}{dt}m_p + \gamma m_p = r, \quad (7)$$

giving

$$m_p = \frac{r}{\gamma}. \quad (8)$$

Putting this together, we get

$$m(t) = m_h(t) + m_p = Ae^{-\gamma t} + \frac{r}{\gamma}. \quad (9)$$

Evaluating at $t = 0$, we get

$$m(t) = m_h(t) + m_p = -(e/\gamma - m_0)e^{-\gamma t} + \frac{r}{\gamma}. \quad (10)$$

The limit $t \rightarrow 0$, is $m(0) = m_0$, which is the initial condition. And the limit $t \rightarrow \infty$ is r/γ the steady state value representing an equilibrium between production and degradation.

(b) Write a code in Python that uses the update rule in its simplest form shown in Figure ?? to find the number of mRNA as a function of time. For concreteness, let's take typical numbers such as that the degradation rate is $\gamma = 1/3 \text{ min}^{-1}$ and $r = 1/3 \text{ min}^{-1}$. Feel free to use your chatbot of choice to help you with the coding, but make sure you use Rob's rules for code produced in this context. We are NOT going for efficiency. I don't want you using a differential equation integrator, I want to do the truly naive Euler forward integration shown in the rate equation protocol. The goal is to make a plot of $m(t)$. Make sure you submit your code with your homework. The code must be commented to demonstrate that you know precisely what is going on. This should be a very simple code. Comment on how long you had to run your numerical integration to reach saturation in terms of the parameters r and γ .

```

import numpy as np
import matplotlib.pyplot as plt

# Parameters
r = 1/3      # production rate, per minute
gamma = 1/2  # degradation rate, per minute
# Steady state
m_ss = r / gamma
dt = 0.01   # minutes
tf = 5/gamma
m0 = 0.0    # initial condition, no mRNA
# store the first values
t_list = [0.0]
m_list = [m0]
t = 0.0
m = m0
# number of time steps up to final time tf
Nsteps = int(tf / dt)

# iterate with update rule
for i in range(Nsteps):
    # forward Euler step:
    m_new = m + dt * (r - gamma * m)
    t_new = t + dt
    # store results
    t_list.append(t_new)
    m_list.append(m_new)
    # update state variables for next iteration
    t = t_new
    m = m_new

# Convert lists to arrays
t_arr = np.array(t_list)
m_arr = np.array(m_list)

# Plot m(t)
plt.figure(figsize=(6, 4))
plt.plot(t_arr, m_arr, lw=2, label="Euler integration")
plt.axhline(m_ss, color="k", ls="--", lw=1.5, label=r"steady state  $m_{\mathrm{ss}}=r/\gamma$ ")
plt.xlabel("time t (min)")
plt.ylabel("mean mRNA m(t)")
plt.legend(frameon=False)
plt.tight_layout()
plt.show()

```

Figure 1: Code

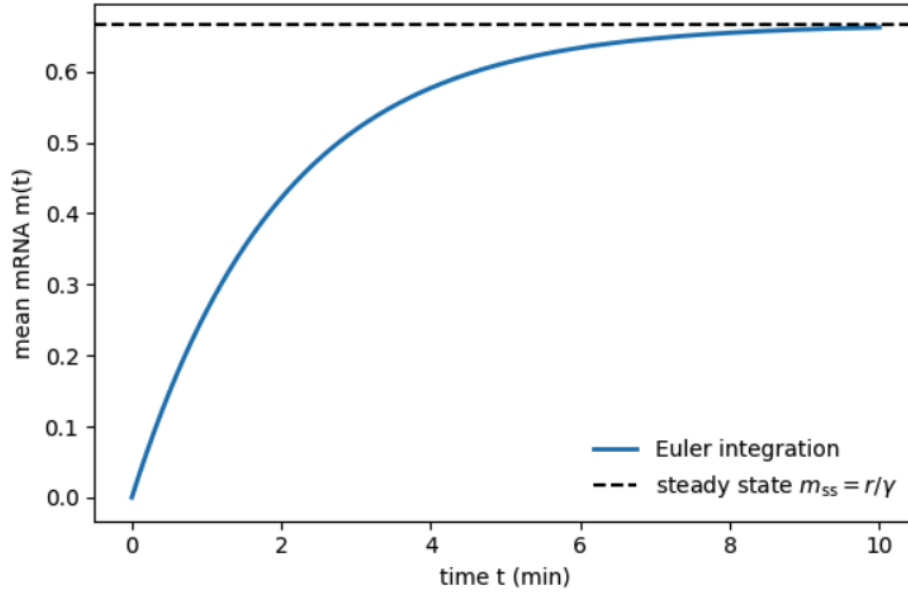


Figure 2: $m(t)$

The time needed to reach saturation is a few $1/\gamma$.

(c) As we have already discussed in class a number of times, most genes are regulated. This means that unlike the constitutive promoter, there is feedback. The simplest example of that behavior is perhaps the auto-activation switch shown in Figure ???. The simplest model we can write of this equation is to imagine that the promoter produces mRNA with a higher rate r_∞ when it is self-activated. We write an equation of this form as

$$\frac{dA}{dt} = -\gamma A + r_0 \frac{1}{1 + \left(\frac{A}{K}\right)^2} + r_\infty \frac{\left(\frac{A}{K}\right)^2}{1 + \left(\frac{A}{K}\right)^2}. \quad (11)$$

Sketch a diagram of the rates vs A as I have done in class. In particular, your graph should show the degradation term and then the production term and beneath it put a phase portrait that shows which direction the vectors are pointing. Using your diagram, make statements about the fixed point solutions of this equation and comment on their stability. Does this make sense?

Solutions:

We have

$$\frac{dA}{dt} = -g(A) + f(A), \quad (12)$$

with

$$g(A) = \gamma A \quad (13)$$

and

$$f(A) = r_0 \frac{1}{1 + \left(\frac{A}{K}\right)^2} + r_\infty \frac{\left(\frac{A}{K}\right)^2}{1 + \left(\frac{A}{K}\right)^2}. \quad (14)$$

Steady state happens when

$$\frac{dA}{dt} \Big|_{A=A_{ss}} = 0. \quad (15)$$

Which give us

$$g(A_{ss}) = f(A_{ss}). \quad (16)$$

This is an activating system so $r_\infty > r_0$. We can plot the two curves to find their intersections.

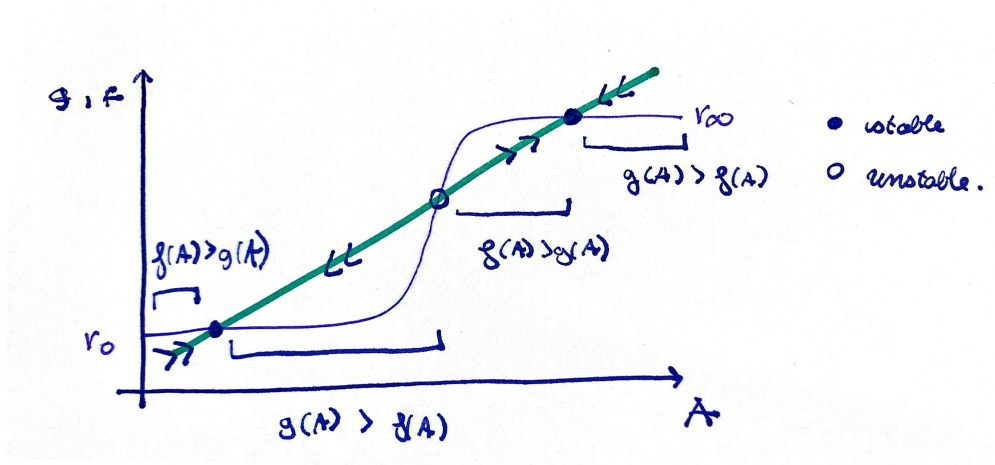


Figure 3: Stability and steady state

How can we differentiate stable fixed points from unstable fixed points?

We write $A = A_{ss} + \delta A$, with small δA , to look at what's going on around a fixed point. We write

$$\frac{dA_{ss}}{dt} + \frac{d\delta A}{dt} = -g(A_{ss} + \delta A) + f(A_{ss} + \delta A). \quad (17)$$

and $\frac{dA_{ss}}{dt} = 0$, so

$$\frac{d\delta A}{dt} = -g(A_{ss} + \delta A) + f(A_{ss} + \delta A). \quad (18)$$

Therefore we see if $g(A_{ss} + \delta A) > f(A_{ss} + \delta A)$ (i.e. the green curve is above the purple curve) $\frac{d\delta A}{dt} < 0$ and the perturbation decreases to 0 and the system goes back to the steady state, stable fixed point. Otherwise the perturbation grows and the system goes away from the steady state, unstable fixed point.

At low A , the promoter is mostly “off,” so production is close to the basal rate r_0 , while degradation is small but grows linearly with A . If r_0 is not large enough to balance degradation at higher A , the system can sit in a low- A stable state.

As A increases past the threshold set by K , auto-activation turns on and the production is closer to r_∞ , which can create high- A stable state. In that bistable regime, the intermediate fixed point is unstable and acts as a threshold: small perturbations decay back to the low state, while sufficiently large increases in A push the system to high A steady state.

(d) Use your code from part (b) but now modified to account for the more complex right hand side to integrate the equations of motion for the auto-activation switch. Let’s keep the same value of γ , and consider $r_0 = 1/10 \text{ min}^{-1}$ and $r_\infty = 5 \text{ min}^{-1}$. We also need an equilibrium constant and in my dimensionless units, consider $K = 5$. Solve for $A(t)$ for several different initial conditions to show that for low initial A , the solution converges to the small A fixed point and for large initial A , the solution converges to the high A fixed point.

```

# Parameters
gamma = 1/3          # min-1
r0 = 1/10            # min-1
rinf = 5             # min-1
K = 5                # mol number
dt = 0.01            # min
tf = 20/gamma        # final time
# production term for auto-activation
def production(A):
    x = (A / K)**2
    return r0 * (1/(1 + x)) + rinf * (x/(1 + x))
# right-hand side dA/dt
def f(A):
    return -gamma * A + production(A)
# several initial conditions
A0_list = [0.0, 0.5, 1.0, 3.0, 6.0, 12.0]
plt.figure(figsize=(6, 4))
for A0 in A0_list:
    # store the first values
    t_list = [0.0]
    A_list = [A0]
    t = 0.0
    A = A0
    Nsteps = int(tf / dt)
    # iterate with update rule (forward Euler)
    for i in range(Nsteps):
        A_new = A + dt * f(A)
        t_new = t + dt
        t_list.append(t_new)
        A_list.append(A_new)
        t = t_new
        A = A_new
    t_arr = np.array(t_list)
    A_arr = np.array(A_list)
    plt.plot(t_arr, A_arr, lw=2, label=fr"$A_0={A0}$")
plt.xlabel("time t (min)")
plt.ylabel("A(t)")
plt.legend(frameon=False, fontsize=9)
plt.tight_layout()
plt.show()

```

Figure 4: Code

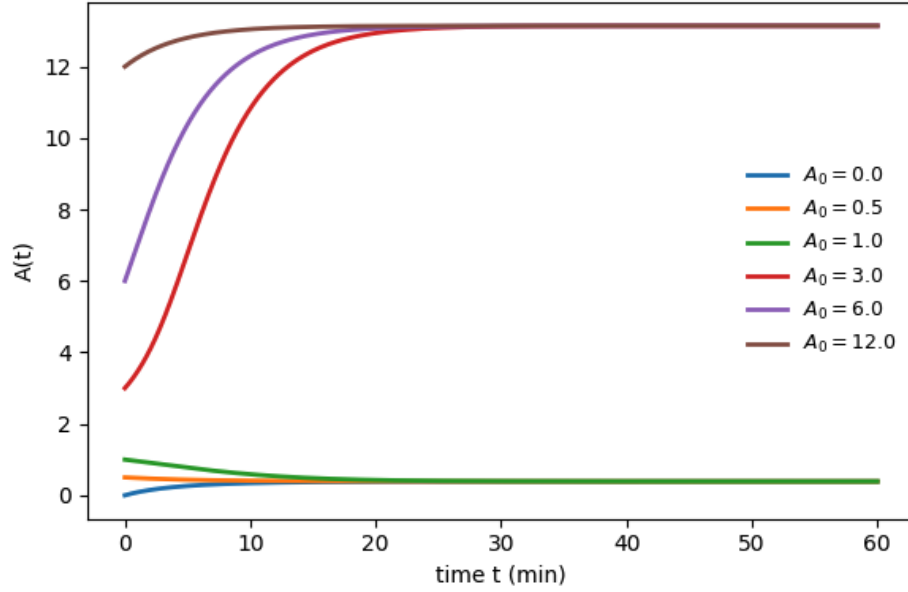


Figure 5: $A(t)$ for different A_0

4. Equation of Motion for Mean Cytoskeletal Filament Length

In class we discussed the rate equation protocol shown in Figure ???. Our application of the protocol in class was to the problem of a constitutive promoter and provided a dynamical equation for the average number of mRNAs per cell as a function of time. In this problem, you are going to imitate that analysis, but this time thinking about the average length of a cytoskeletal filament as a function of time. Imagine a situation in which we have a closed box in which a single cytoskeletal filament has been nucleated (using a nucleating factor, for example) and which is bathed in a reservoir of monomers, with the initial number of monomers being given by N_{tot} . Our goal is to compute $L(t)$, where L is the length of the filament as a function of time. The rate at which monomers attach is $k_{on}n_{free}$, where n_{free} is the number of free monomers and the rate at which monomers detach from the tip of the growing filament is k_{off} . Write a dynamical growth equation for the dynamics of $L(t)$ and find the solution. What is the steady-state length of the filament? Make a plot of the length as a function of time - you can attempt to figure out reasonable choices of the parameters by looking at book.bionumbers.org or by looking at PBoC, but give an explanation of your choices. Also, compare and contrast the analysis here with that done in class for the constitutive promoter.

Solution:

Let $n(t)$, be the total number of monomers in the filament. Each monomer has length a . $L(t) = an(t)$. The number of free monomer is $n_{free}(t) = N_{tot} - n(t)$. The growth equation is

$$\frac{dn}{dt} = k_{on}(N_{tot} - n) - k_{off}. \quad (19)$$

The steady state for L , is

$$L_{ss} = a(N_{tot} - k_{off}/k_{on}). \quad (20)$$

With a similar approach that 3a), we get the full time dependent solution

$$L(t) = L_{ss} + (L_0 - L_{ss})e^{-k_{on}t} \quad (21)$$

The approach to the steady state is set by $1/k_{on}$. We take $k_{off} \approx 1s^{-1}$, and $k_{on} \approx 10^{-2}s^{-1}$. These values are taken from Kuhn, Jeffrey R., and Thomas D. Pollard. "Real-time measurements of actin filament polymerization by total internal reflection fluorescence microscopy." *Biophysical journal* 88.2 (2005): 1387-1402. In the paper, they give $k_{on} \approx 10\mu M^{-1}s^{-1}$, and we remember from previous homework that 1 molecule in a bacteria cell of a volume $1\mu m^3$, corresponds to a concentration of 1 nM, we multiplied the value of the paper by 10^{-3} . We take the unit a to be around a few nano meters.

```

Ntot = 2000          # total monomers in the closed box
koff = 1             # per sec
kon = 1e-2           # per sec
a = 3                # nm
dt = 0.01            # sec
tf = 500             # sec (should be several times 1/kon = 100 s)
# initial condition
n0 = 0.0
def dn_dt(n):
    n_free = Ntot - n
    return kon * n_free - koff
# storing
t_list = [0.0]
n_list = [n0]
t = 0.0
n = n0
Nsteps = int(tf / dt)
for i in range(Nsteps):
    n_new = n + dt * dn_dt(n)
    t_new = t + dt
    # keep n in [0, Ntot]
    n_new = min(max(n_new, 0.0), float(Ntot))
    t_list.append(t_new)
    n_list.append(n_new)
    t, n = t_new, n_new
t_arr = np.array(t_list)
L_arr = a * np.array(n_list) # filament length in nm
# ss
n_ss = max(Ntot - koff / kon, 0.0)
L_ss = a * n_ss
# plot
plt.figure(figsize=(6, 4))
plt.plot(t_arr/60, L_arr/10**3, lw=2, label="Euler integration")
plt.axhline(L_ss/10**3, color="k", ls="--", lw=1.5, label="steady state")
plt.xlabel("time t (min)")
plt.ylabel("filament length L(t) (micro m)")
plt.legend(frameon=False)
plt.tight_layout()
plt.show()

```

Figure 6: Code

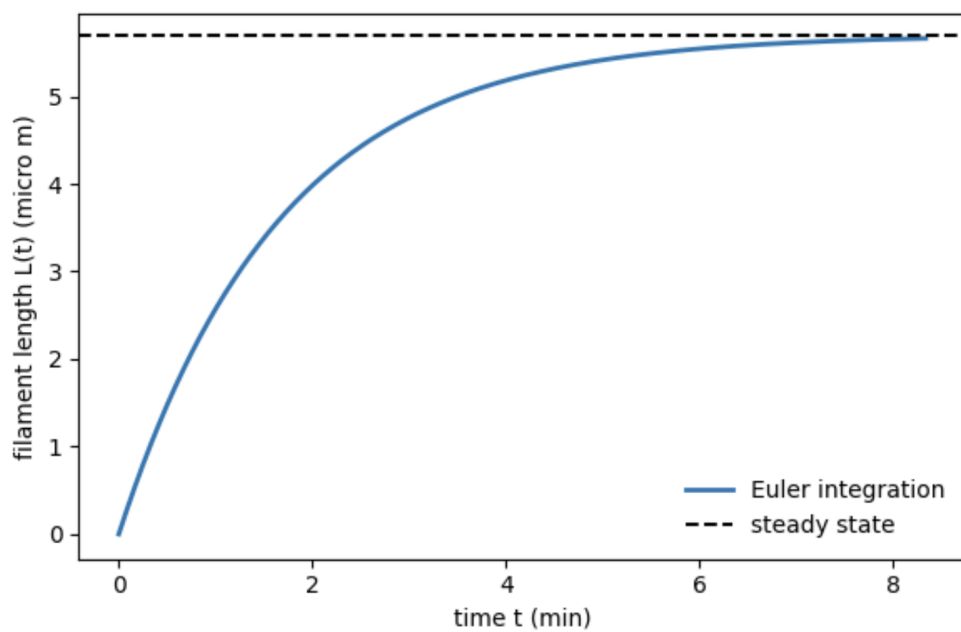


Figure 7: $L(t)$ in μm .

Note: if the Student considered a length in number of monomers, give the full points.